DSA5208: Scalable Distributed Computing for Data Science

Lecture 0: Overview

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About the lecture

About the lecturers:

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Assessment:

- Three projects (30%, 35%, 35%)
- No midterm or final exam!

1 Motivation and challenges

2 Topics in this course (First 6 weeks)

Data never sleeps: AI Edition 2025

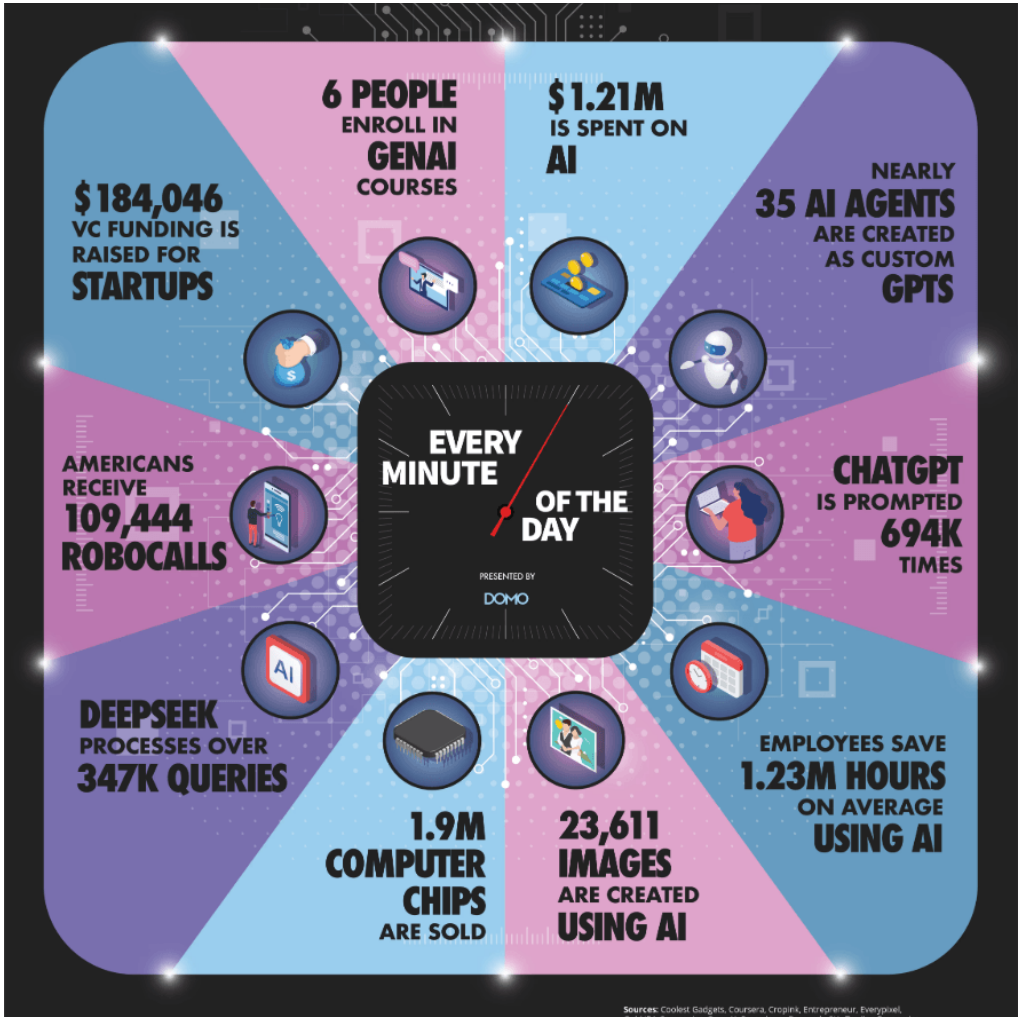
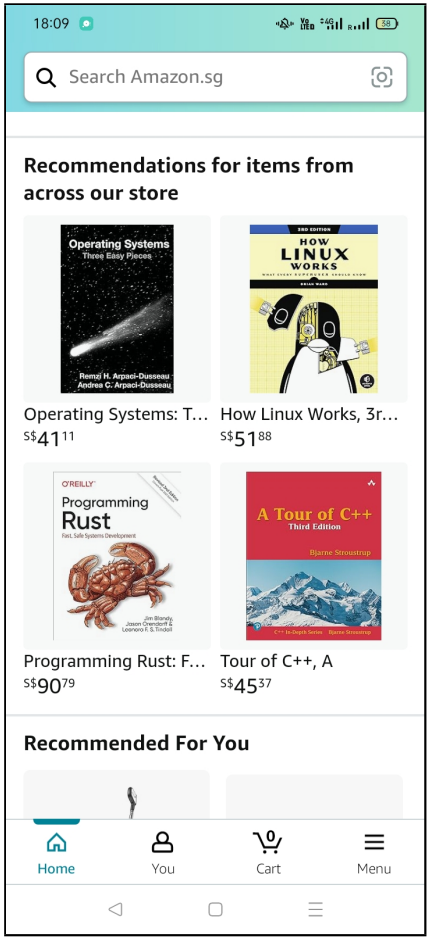


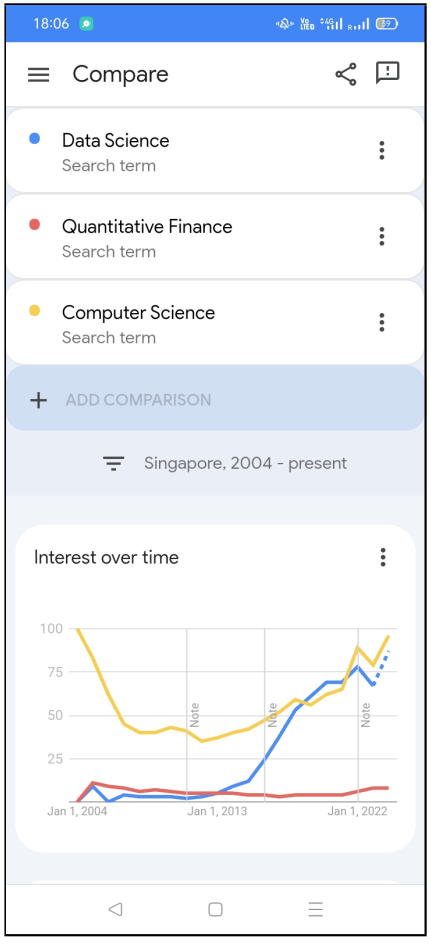
Figure from `https://www.domo.com/learn/infographic/data-never-sleeps-ai-edition-2025`

Applications of big data

- Product Recommendation (Amazon)

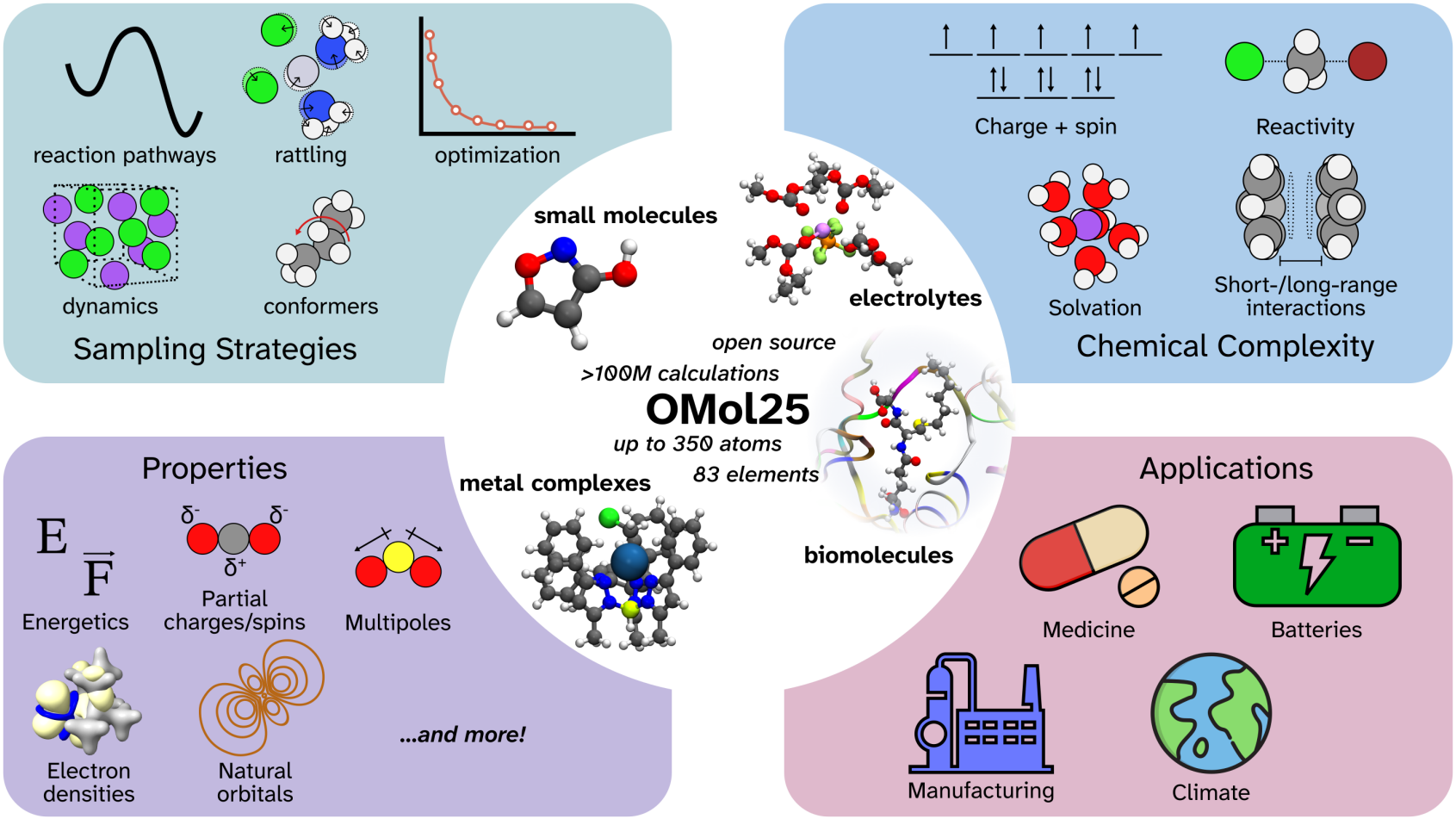


- Google Trends (Google)



Applications of big data

Meta's OMol25 dataset:



Our machines

- **RAM (Random-access memory):**
  - Size: 8GB to 256GB
  - Speed: DDR4 between 2133 and 5000 MT/s, DDR5 between 4800 and 9600 MT/s, DDR6 between 8800 and 17600 MT/s.
- **Hard disk drives:**
  - Size: 500GB to 30TB+
  - Access time: 5ms to 15ms
  - Data transfer rate: 80MB/s to 300MB/s
- **Solid-state drives:**
  - Size: 128GB to 4TB
  - Access time: 20μs to 100μs
  - Data transfer rate: 500MB/s to 14000MB/s

Hardware will fail

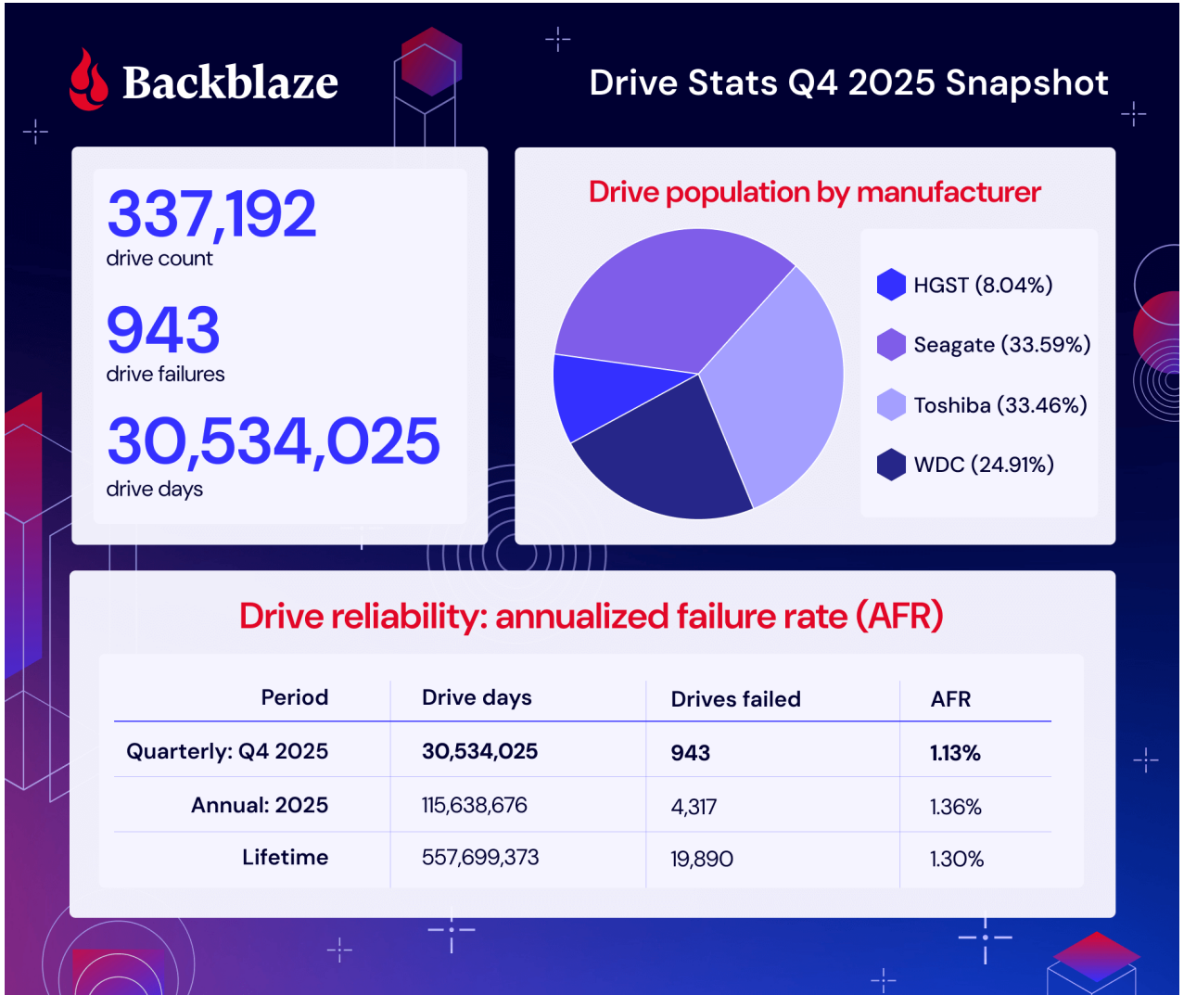


Figure from https://www.backblaze.com/blog/backblaze-drive-stats-for-2025/

Everything can go wrong...

Device	Proportion
HDD	81.84%
Miscellaneous	10.20%
Memory	3.06%
Power	1.74%
RAID card	1.23%
Flash card	0.67%
Motherboard	0.57%
SSD	0.31%
Fan	0.19%
HDD backboard	0.14%
CPU	0.04%

Table: Failure percentage from over 290,000 reports

Data from "What Can We Learn from Four Years of Data Center Hardware Failures?" by Wang, Xu, and Zhang

Challenges in designing distributed algorithms

- Synchronization and coordination among processes
- Data storage and access
- Consistency and replication
- Fault tolerance
- Scalability and modularity
- ...

1 Motivation and challenges

2 Topics in this course (First 6 weeks)

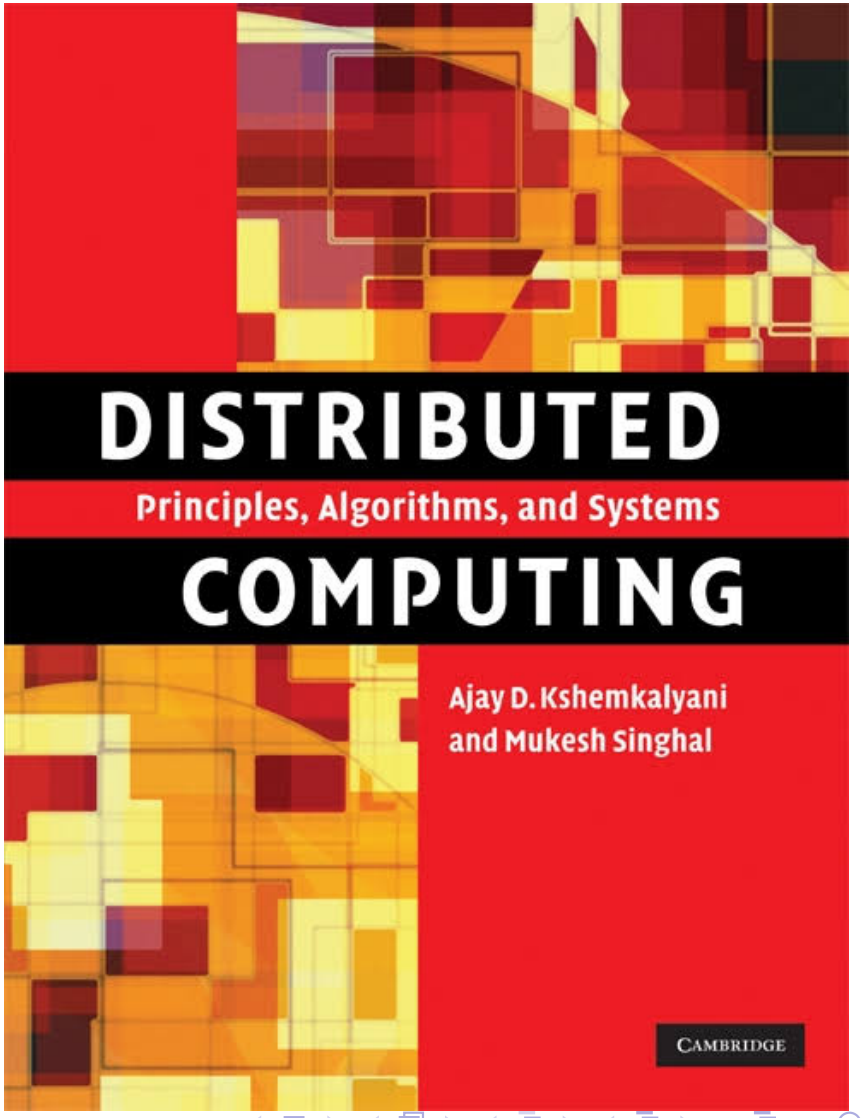
Topics in this course (First 6 weeks)

- **Lecture 1:**

  Physical and Logical Times

- **Lecture 2:**

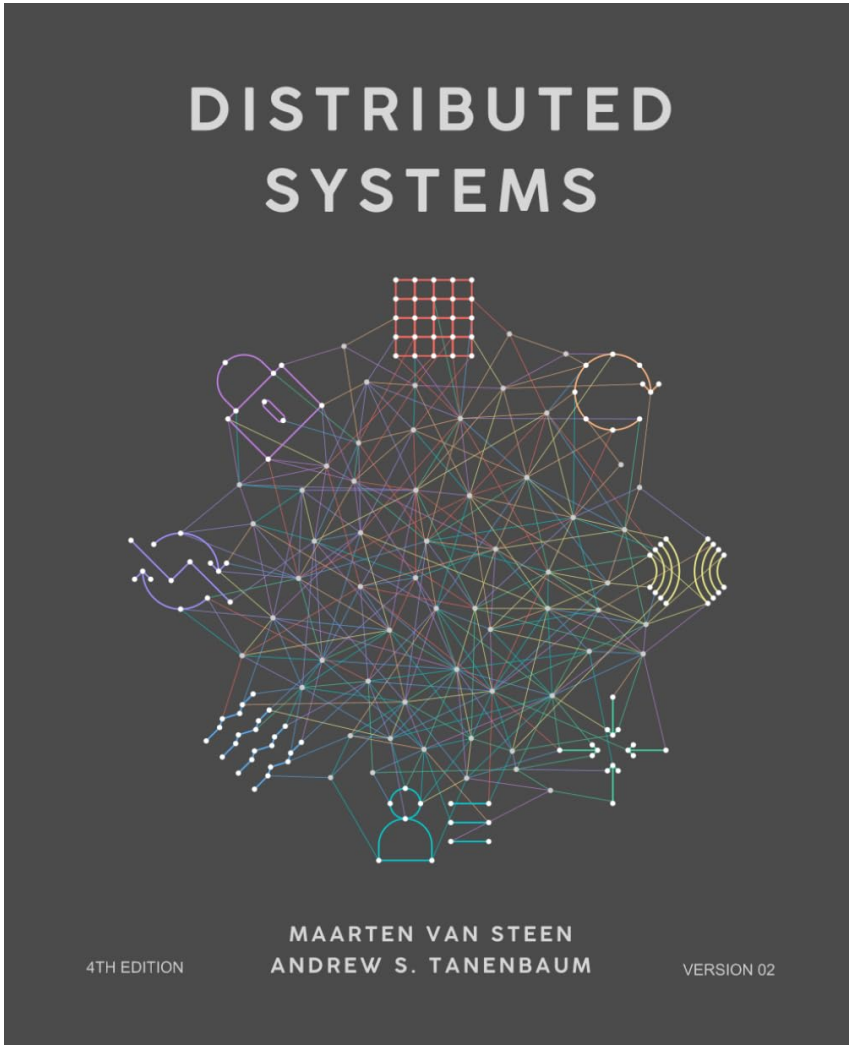
  Broadcast and Synchronizers



Topics in this course (First 6 weeks)

- **Lecture 3:**

  Consistency Models



Topics in this course (First 6 weeks)

- **Lecture 4:**

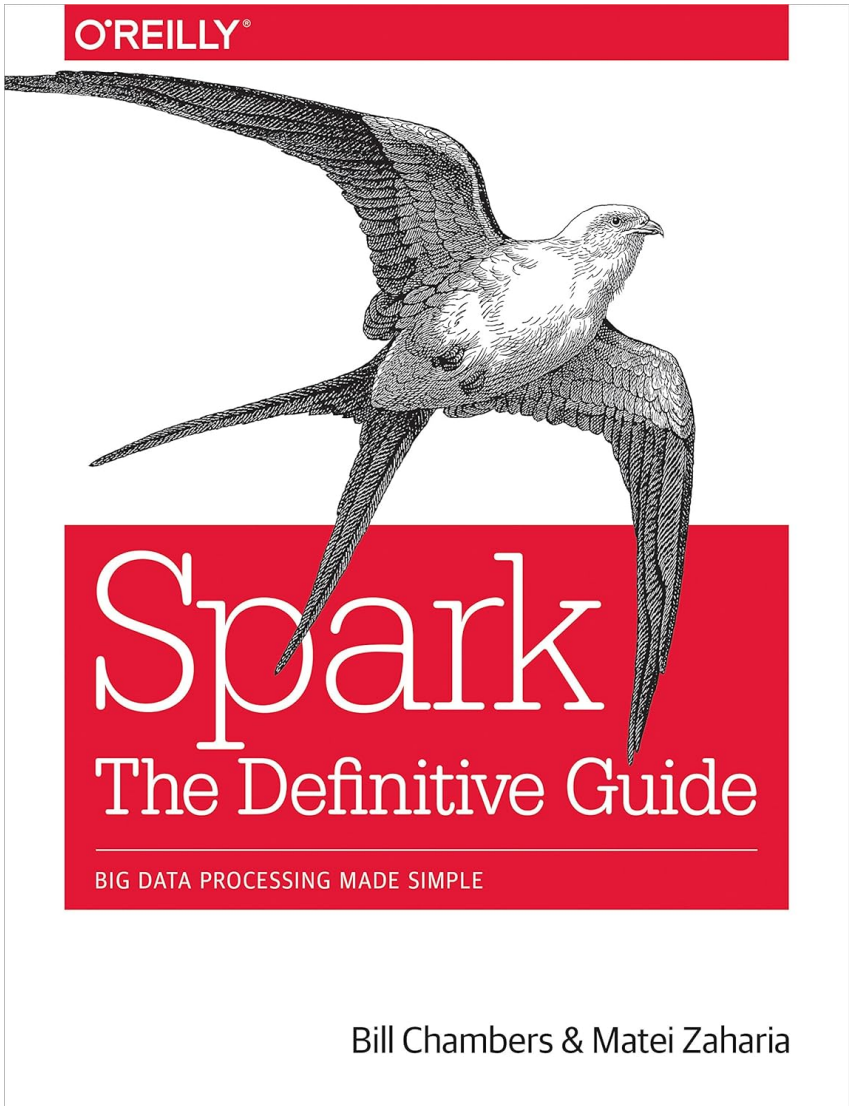
  Introduction to Apache Spark

- **Lecture 5:**[a]

  Distributed Algorithms in Apache Spark

[a] This lecture is mainly based on the source code of Spark and some online documents:

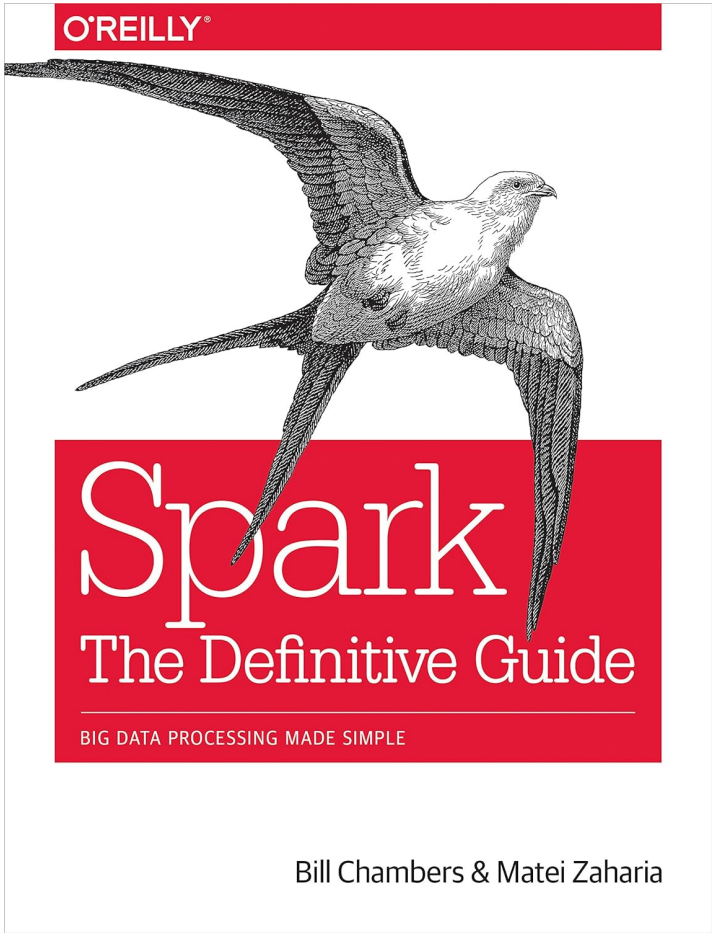
- Link 1
- Link 2
- Link 3
- Link 4
- Link 5

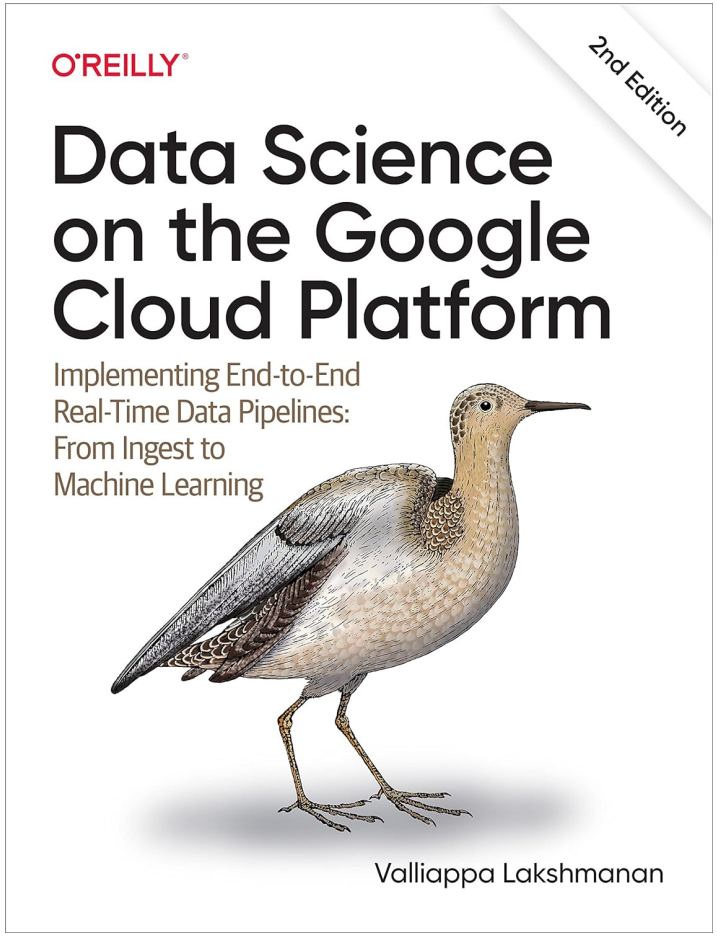


Topics in this course (First 6 weeks)

- **Lecture 6:**

  Scalable Machine Learning with MLlib and Cloud Computing





First two projects

Group projects: (at most 3 people per group)

1. Installation of distributed database systems and tests on consistency models.
2. Use Apache Spark for a machine learning task.